

Cognitive Style Shapes Search Behaviours: An fNIRS Study of Exploratory Search

Huimin Tang
Huimin.Tang@nottingham.edu.cn
School of Computer Science,
University of Nottingham Ningbo
China
Ningbo, China

Boon Giin Lee
BOON-
GIIN.LEE@nottingham.edu.cn
School of Computer Science,
University of Nottingham Ningbo
China
Ningbo, China

Dave Towey
Dave.Towey@nottingham.edu.cn
School of Computer Science,
University of Nottingham Ningbo
China
Ningbo, China

Max L. Wilson
Max.Wilson@nottingham.ac.uk
School of Computer Science,
University of Nottingham
Nottingham, United Kingdom

Matthew Pike*
Matthew.Pike@nottingham.edu.cn
School of Computer Science,
University of Nottingham Ningbo
China
Ningbo, China

Abstract

Cognitive style, a user's habitual approach to information processing, offers a promising approach to personalising information searching (IS) systems, yet the underlying style-related search behaviours remain poorly understood. This study investigates how the Wholist-Analytic and Verbal-Imagery dimensions of cognitive style influence search behaviour and prefrontal cortex (PFC) activation during the Post-focus stage of exploratory search. Forty participants completed comparison search tasks while we recorded behavioural metrics, subjective workload ratings, and functional near-infrared spectroscopy (fNIRS) data. Our results demonstrate that cognitive style significantly predicted search engine results page (SERP) interaction patterns: analytics and imagers adopted structured navigation with detailed reading, whereas wholists and verbalisers preferred sporadic navigation with rapid scanning. Critically, fNIRS revealed distinct PFC activation patterns, specifically in the Ventrolateral PFC (VLPFC) and Dorsolateral PFC (DLPFC), corresponding to these behavioural differences. By mapping neuro-cognitive profiles to IR behaviours, this work provides empirical grounding for designing "style-aware" adaptive IR interfaces that tailor information density and navigational support to individual cognitive profiles.

CCS Concepts

• **Information systems** → **Users and interactive retrieval**; • **Human-centered computing** → **Empirical studies in HCI**; **Web-based interaction**.

*Corresponding author. The data presented in this study are available upon reasonable request from the corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License. *SIGIR '26, Melbourne, VIC, Australia*
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2599-9/2026/07
<https://doi.org/10.1145/3805712.3809541>

Keywords

Cognitive Style; fNIRS; Exploratory Search; Mental Workload; Interactive Information Retrieval;

ACM Reference Format:

Huimin Tang, Boon Giin Lee, Dave Towey, Max L. Wilson, and Matthew Pike. 2026. Cognitive Style Shapes Search Behaviours: An fNIRS Study of Exploratory Search. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '26)*, July 20–24, 2026, Melbourne, VIC, Australia. ACM, New York, NY, USA, 13 pages. <https://doi.org/10.1145/3805712.3809541>

1 Introduction

Users increasingly expect search systems to adapt to their individual needs. The traditional "one-size-fits-all" approach fails to account for individual differences [20, 59]. This limitation becomes particularly acute in complex exploratory tasks, where modern systems—including those augmented by large language models (LLMs)—must support diverse user strategies and preferences [68, 78].

While the information retrieval (IR) community has long sought to move beyond a singular design for all users, current personalisation methods typically require systems to learn from extended interaction histories or demand that users explicitly specify their preferences [55, 78]. These reactive approaches delay the benefits of personalisation until sufficient data accumulates, potentially imposing unnecessary mental workload (MWL) on users who must repeatedly adapt to interfaces misaligned with their natural information processing tendencies [21, 43]. An alternative approach leverages users' stable, intrinsic traits to proactively configure IS systems from the first interaction, aligning system behaviour with individual preferences before any interaction occurs [21].

Research confirms that stable user traits significantly influence preferences and behaviours in IS systems [74], and that tailoring designs to these traits demonstrably enhances usability [60, 76]. Personality traits such as the "Big Five" [41], describing patterns of thinking, feeling, and behaviour, are associated with search planning [4] and interface navigation [56]. Cognitive style—a user's

habitual way of processing information [67]—similarly influences query formulation, web navigation, and information processing [50, 60, 103].

Despite substantial behavioural evidence linking user traits to IS behaviour, the underlying cognitive processes driving these behaviours remain poorly understood [10, 66]. This mechanistic understanding is essential for generalising findings across contexts and applying them confidently to system design [21]. Although initial studies have examined the neural correlates of personality traits [32], cognitive style has received comparatively little attention [10]. Existing behavioural findings are notably inconsistent, with studies reporting contradictory navigation and processing strategies [33, 46, 50, 103] (discussed in Sec. 2.2). These contradictions may reflect uncontrolled variation in task difficulty or search stage, and highlight that observable behaviours alone cannot reveal the cognitive mechanisms involved. Researchers have consequently called for neurophysiological methods to investigate these underlying processes [10]. While neuroimaging studies demonstrate that cognitive styles recruit distinct brain regions during controlled laboratory tasks [53, 91], and initial fNIRS work has examined search behaviour [103] (discussed in Sec. 2.3), the neurophysiological basis of cognitive style differences during realistic IS tasks remains unexplored.

To address this gap, we present an investigation into the cognitive processes underlying users' cognitive styles (Wholist-Analytic and Verbal-Imagery dimensions [87]) during the *Post-focus* stage of exploratory search [105], where users shift from broad exploration to targeted information gathering and evaluation. This study addresses two research questions:

RQ1 How does cognitive style influence IS behaviours during the *Post-focus* stage of exploratory search?

RQ2 How do observable behavioural differences relate to underlying patterns of brain activity?

Participants completed two exploratory search tasks in a controlled laboratory setting. IS behaviours were captured through screen recordings and think-aloud protocols, while functional near-infrared spectroscopy (fNIRS) measured brain activation in the prefrontal cortex (PFC)—a region central to executive function and cognitive workload [26, 34]. This multimodal approach enables us to move beyond self-report and behavioural metrics to examine how cognitive style shapes both observable behaviour and underlying brain activity.

Our findings make two key contributions:

- **Behavioural Differences:** Cognitive style significantly predicts search behaviour, with distinct strategies for query formulation and information collection emerging during the *Post-focus* stage.
- **Brain Activation Patterns:** fNIRS data reveal that these behavioural differences correspond to distinct patterns of PFC activation, providing neurophysiological evidence for the cognitive processes underlying style-related behaviours.

These findings establish a foundational understanding of why users with different cognitive styles exhibit distinct search behaviours, with direct implications for the design of adaptive search interfaces. Our work further demonstrates that fNIRS can reliably capture cognitive processes during dynamic real-world IS tasks,

enabling future research linking neurophysiology to complex information behaviour.

2 Related Work

This section reviews cognitive style as a stable user trait (Sec. 2.1), its documented influence on IS behaviours (Sec. 2.2), and the current understanding of the cognitive and brain activation patterns underlying these behaviours (Sec. 2.3).

2.1 Cognitive Style

Cognitive style refers to an individual's habitual way of perceiving, organising, and processing information [67], a construct studied since the 1940s [31]. Crucially for IS personalisation, cognitive style is considered stable and consistent across different environments and tasks [19]. Although some researchers suggest that cognitive style may adapt to fundamental environmental changes [52], substantial evidence demonstrates that individuals perform worse when required to process information in ways misaligned with their inherent style [97, 109]. This stability distinguishes cognitive style from related concepts such as "learning style" or "decision-making style," which may be influenced by training or environmental factors [1, 57], and makes it particularly suitable for personalised systems.

Riding and Cheema [87] reorganised earlier cognitive style dimensions into two orthogonal, fundamental dimensions: Wholist-Analytic (WA) and Verbal-Imagery (VI) (Fig. 1). This reorganisation addressed the poor reliability of previous models stemming from unclear definitions and limited replication [35]. The two dimensions are independent [89]: the WA dimension classifies individuals by their preference for processing information holistically (wholist) or by focusing on component parts (analytic), while the VI dimension classifies individuals by their preferred representational modality—verbal (favouring textual information) or imagery (favouring visual information) [87]. This model provides a robust foundation for contemporary cognitive style research [15, 51].

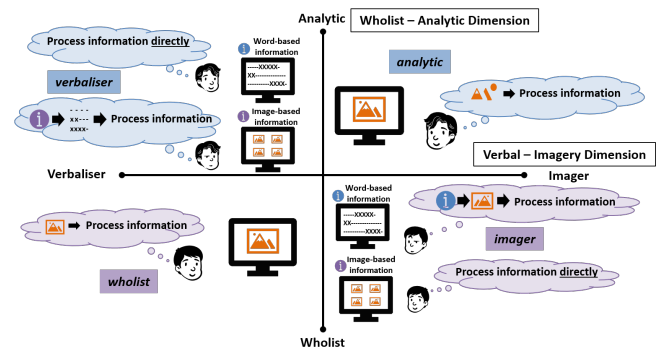


Figure 1: Cognitive style dimensions (adapted from Riding and Cheema [87])

2.2 Cognitive Style and Information Searching

Research has established that cognitive style influences IS behaviours, including search strategy, query formulation, and SERP interaction.

Kinley et al. [46–49] identified differences in search strategy and query formulation across cognitive style groups. They observed that verbalisers and wholists preferred a *Top-down* approach (general-to-specific searching), while imagers and analytics preferred a *Bottom-up* strategy (specific keyword expansion). For query refinement, verbalisers and analytics frequently used *Replace*, *Add*, and *Remove* actions, whereas wholists and imagers more often entered *New* queries or *Repeated* previous ones. However, these patterns lack replication, leaving it unclear whether they generalise across the WA dimension.

Kinley et al. [50] further reported distinct SERP navigation and processing styles. Verbalisers adopted a *Sporadic* navigation strategy (frequent reformulation based on initial results), while imagers, wholists, and analytics used a more *Structured* approach. For information processing, verbalisers and analytics typically performed rapid *Scanning*, whereas wholists and imagers preferred thorough *Reading*. Chen et al. [14] reinforced this pattern on the WA dimension, finding that analytic users favoured careful reading of detailed information while wholist users preferred scanning summarised results.

More recently, Tang et al. [103] confirmed Kinley’s VI findings and additionally showed that verbalisers experienced higher MWL than imagers when using a standard Google interface. However, the literature remains inconsistent: other studies report that verbalisers tend to structure their search using the *Bottom-up* approach, *Structured* navigation, and *Reading* strategies [28, 33], with imagers showing the opposite pattern.

These contradictions likely arise from methodological differences—particularly uncontrolled variation in task complexity and search stage—and underscore the limitations of relying solely on observable behaviour to understand cognitive style effects. These limitations motivate RQ1, which examines behavioural differences during a controlled *Post-focus* task.

2.3 Cognitive Processes and Mechanisms

Understanding the cognitive processes underlying IS behaviour requires both theoretical frameworks and empirical methods capable of capturing internal mental states.

Vakkari’s three-stage Information Search Process (ISP) model [105], adapted from Kuhlthau [54], characterises exploratory search as progressing through distinct phases. The *Pre-focus* stage involves initiating and exploring broad information needs. The *Focus Formulation* stage establishes a specific perspective. The *Post-focus* stage shifts to targeted information gathering and evaluation.

At the cognitive level, Sutcliffe and Ennis [99] proposed a cyclical model comprising four activities: *Problem identification*, *Need articulation* (retrieving relevant terms), *Query formulation*, and *Result evaluation*. Users iterate through these activities within each ISP stage, and this framework enables analysis of how behaviours differ across stages [99].

A growing body of IS neuroimaging work demonstrates that these activities carry identifiable neural signatures. Electroencephalography (EEG) and functional magnetic resonance imaging (fMRI) studies have investigated neural correlates of specific IS activities, including query formulation [30, 40, 95], query submission [40], result viewing [30, 40, 45, 95], mouse-click behaviour [108], content

page reading [30, 40, 45, 95], and page bookmarking [30]. Further work has identified neural correlates of information need, linking moments of uncertainty during search to activation in memory and decision-related regions [70, 71]. Overall, these findings establish that neuroimaging methods are sensitive to the kinds of cognitive distinctions that arise during realistic search activity.

However, there is a significant gap in the mechanistic understanding of how cognitive style influences these processes [10]. Miller’s Information Processing Model [69] theoretically linked the WA dimension to perception and the VI dimension to memory during an information processing task, which would happen frequently during all IS activities, but this framework remains empirically untested [104]. Researchers have consequently advocated for physiological methods to investigate underlying cognitive functions [10]. Early neuroimaging work using EEG [90, 91] and fMRI [53] demonstrated that individuals with different cognitive styles recruit distinct brain regions when converting information into their preferred format. However, these studies employed highly controlled laboratory tasks far removed from realistic IS contexts, and both EEG and fMRI are sensitive to body movements [102], constraining the ecological validity necessary for studying dynamic search behaviour [52].

fNIRS offers a viable alternative, providing tolerance for movement during interactive tasks while maintaining sensitivity to cognitive processes in the PFC [12, 34]. Yet fNIRS research linking cognitive style to IS behaviour remains scarce. Tang et al. [103] represents the only prior study, examining the VI dimension during web search and finding differential MWL patterns. However, that work did not investigate the WA dimension, nor did it systematically analyse the brain activation patterns underlying the observed behavioural differences. The present study addresses these gaps: RQ1 examines behavioural differences across both cognitive style dimensions, while RQ2 investigates the corresponding prefrontal activation patterns.

3 Experiment Design

To address the research questions, this study combined behavioural logging (RQ1) with neurophysiological measurement (RQ2). The *Post-focus* stage was selected because it is more experimentally tractable for cognitive measurement [66] and represents the critical transition from exploration to focused collection—where individual differences in information processing should be most apparent, making stable cognitive style differences more discernible here than during initial broad exploration. fNIRS was chosen over EEG and fMRI for its tolerance of body movements and portability during interactive tasks [102], and has proven sensitive to cognitive processes and MWL in the PFC [23, 34] during IS tasks [12, 61].

3.1 Participants

Forty participants (22 male, 17 female, 1 non-binary; all native English speakers), aged 19–45 years ($M = 25.75$, $SD = 5.93$), were recruited from the University of Nottingham. All held or were completing university-level qualifications; 36 were students, and 4 were teaching fellows. All had normal or corrected vision and reported no history of head trauma or brain injury. The study received ethical approval from the university’s ethics committee.

Participants provided informed consent and received a £20 gift voucher. The reported participant IDs are non-sequential due to non-attendance during recruitment.

To minimise the influence of prior domain knowledge on search behaviour [99], participants completed a Task Familiarity questionnaire upon arrival, and the two topics with which they reported least familiarity were assigned as their search tasks. This ensured participants began with minimal pre-existing knowledge, reducing confounding variation in search strategy.

Participants were classified into cognitive style groups using two validated instruments developed by Peterson and colleagues: the Verbal-Imagery Cognitive Style (VICS) test and the Extended Wholist-Analytic (E-WA) test [80, 82]. These instruments are refined versions of the Cognitive Style Analysis [88], developed to address known psychometric limitations [79, 86], and are confirmed to be independent of intelligence and personality [81]. The VICS test measures performance on verbal stimuli (requiring conceptual judgements) and imagery stimuli (requiring comparison of real-world object sizes) [72]. The E-WA test assesses the WA dimension through two tasks: comparing complex geometric figures (wholist processing) and identifying simple figures embedded within complex ones (analytic processing).

3.2 Task

3.2.1 Task Design. The *Post-focus* stage involves a combination of exploratory and lookup behaviours with a defined objective: shifting from broad knowledge acquisition to targeted gathering and evaluation of information [105]. This aligns with borderline exploratory search tasks such as comparison searching [6], which feature a defined path focused on refining understanding. We therefore designed comparison search tasks for this study.

An initial list of ten candidate topics was adapted from prior work [58], selecting domains outside participants' expected daily experience to control for prior knowledge effects. Four assessors with exploratory search experience validated each candidate topic, retaining only those where required information was distributed across at least five distinct webpages and could not be obtained from the first three results of any single query (verified up to July 15th, 2024). Four topics met these criteria:

- T1** Compact dishwashers for small kitchens, comparing three models: Bosch Series 2, Zanussi ZDM17301SA, and Beko DVS05C20W.
- T2** Portable power stations for camping, comparing three models: Jackery Explorer 500, EcoFlow River, and Anker Powerhouse II 400.
- T3** Advanced driver assistance systems (ADAS), comparing three systems: Tesla Autopilot, Ford Co-Pilot360, and BMW Driving Assistant Plus.
- T4** Energy-efficient home heating solutions for UK homes, comparing three technologies: heat pumps, smart thermostats, and underfloor heating.

For each task, participants assumed the role of a journalist writing a feature story and were asked to search for information covering specifications, pricing, reviews, and other relevant aspects before writing a summary paragraph.

3.2.2 Task Structure. To enable precise alignment between fNIRS measurements and specific cognitive activities, the task was decomposed according to Sutcliffe and Ennis's framework [99] discussed in Sec. 2.3. This decomposition slows the search process to accommodate the latency of the haemodynamic response measured by fNIRS, which peaks approximately 5–7 seconds after cortical activation onset [106]. Without such separation, haemodynamic signals from successive cognitive activities overlap, obscuring interpretation. This approach has been validated in prior neuroimaging-IS studies [61, 73].

Since *Problem Identification*, *Need Articulation*, and *Query Formulation* are all reflected in query behaviour [100], we structured tasks around two primary activities: *Query Formulation* and *Result Evaluation*. Each task began with initial query formulation (Step A.a), followed by iterative cycles of result evaluation (Step B) and query refinement (Step A.b). The task concluded with participants writing a summary paragraph (Step C), satisfying the completion criteria for a *Post-focus* task (Fig. 2).

To validate engagement and confirm meaningful information acquisition, participants wrote a paragraph summarising their pre-existing topic knowledge before searching. The difference between pre-search and post-search (Step C) outputs confirmed that participants actively collected and integrated new information. Both writing tasks were limited to five minutes.

Search duration was capped at 15 minutes per task, following recommendations for exploratory search studies [6], after which participants were automatically directed to Step C. Participants could conclude earlier via a "Finish" button (available after Step A.a).

To support *Query Formulation* given participants' low domain knowledge, keyword hints listing the target items (e.g., the three product names) were displayed during Steps A.a and A.b. The current query was shown during *Result Evaluation* (Step B), and the previous query appeared as a reference during subsequent formulation steps (Step A.b). To maintain focus on query refinement, SERPs opened in Step B were automatically closed upon transition to Step A.b. Google's autocomplete suggestions were disabled in Step A, and Google's AI-generated summaries were disabled via the `uom=14` URL parameter. Participants were prohibited from using LLM-based tools (e.g., ChatGPT) during search, as such tools introduce personalisation effects that would confound our manipulation.

3.3 Measures

3.3.1 Search Behaviours. Assessing exploratory search task outcomes is challenging due to their open-ended nature [44], and no consensus exists on universal success metrics [66]. This study, therefore, focused on four behavioural measures (Table 1) adapted from Kinley et al. [50], integrating established subjective and quantitative metrics, directly addressing RQ1.

3.3.2 Neurophysiological Activation. fNIRS data were collected using the Brite headband (Artinis Medical Systems; continuous-wave fNIRS device)¹ and recorded with OxySoft software (Artinis Medical Systems)². The device uses 760 and 850 nm wavelengths and was

¹Artinis Brite: <https://www.artinis.com/brite-frontal>

²Artinis OxySoft: <https://www.artinis.com/oxysoft>

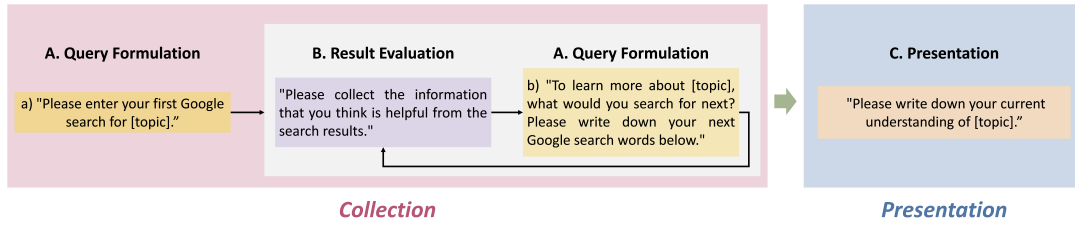


Figure 2: Search task structure showing the iterative cycle of Query Formulation (Step A) and Result Evaluation (Step B), concluding with a summary paragraph (Step C). Rest periods between steps accommodate fNIRS haemodynamic response latency.

Search Behaviour	Measure Metrics
Query Formulation Behaviour	Evaluated by comparing queries entered across steps. Participants could exhibit multiple formulation behaviours (e.g., <i>Add, Delete, Replace, New, Repeat</i>).
Information Search Strategy	Derived from recorded queries and the content analysis [24] of the concurrent think-aloud logs [75], which reveal real-time cognitive processes without affecting task performance or fNIRS measurements [83]. Strategies were classified by how participants' search scope changed over time (<i>Top-down, Bottom-up, or No change</i>).
Web Navigation Style	Inferred from screen recordings and the content analysis [24] of think-aloud logs. Participants were categorised as <i>Sporadic</i> (frequently opening new pages and switching between them) or <i>Structured</i> (systematic progression through results).
Information Processing Approach	Drawing on think-aloud content analysis [24] and screen recordings, participants were categorised as <i>Scanning</i> (rapid assessment of multiple items) or <i>Reading</i> (thorough examination of individual items).

Table 1: Search behaviour measures and operationalisations, adapted from Kinley et al. [50]. Each behaviour was derived from screen recordings, query logs, and think-aloud logs.

positioned over the PFC according to the International 10–20 system [39]. The configuration comprised eleven emitters and seven detectors, yielding 23 channels: 20 long-distance channels (35 mm inter-optode distance) for cortical signals and 3 short-separation channels (10 mm) to capture extracerebral signals. Short-channel signals were subsequently regressed from long-channel data to reduce systemic physiological noise [13]. Data were sampled at 50 Hz. These measures address RQ2 by capturing brain activation associated with search behaviours.

3.3.3 Subjective Workload. Subjective MWL was assessed using Instantaneous Self-Assessment (ISA) [42], a validated single-item measure in which participants rate perceived workload from 1 (low) to 5 (high) at fixed intervals. ISA was selected for its minimal disruption to the task flow.

3.4 Procedure

The experiment was conducted in an office-like laboratory (Fig. 3). Participants and the researcher were separated by a dividing screen to minimise interference. Participants used two monitors, a keyboard, and a mouse; both monitors mirrored the researcher's control laptop, enabling real-time observation of search behaviour and experiment logging. The left monitor displayed task instructions and response fields; the right monitor showed SERPs during Step B. Query entry was restricted to Step A on the left monitor.

Task presentation and data collection were controlled via PsychoPy (Python-based experiment software)³ [77]. Event markers were transmitted to the researcher's laptop via Lab Streaming Layer



Figure 3: Laboratory setup. Participants used two monitors (left: task instructions and query entry; right: SERP display) while wearing the fNIRS headband. A dividing screen separated participants from the researcher, whose mirrored display enabled real-time observation.

(LSL; synchronisation middleware)⁴ [98], ensuring precise temporal alignment with fNIRS recordings. A tablet on the researcher's side monitored fNIRS signal quality in real time. Task order and topic assignment followed a Latin square design to counterbalance learning and order effects [5].

Upon arrival, participants were briefed on the study and provided informed consent. They then completed the Task Familiarity questionnaire and the cognitive style tests (VICS and E-WA), taking approximately 25–30 minutes. Following the cognitive style tests, the researcher reviewed familiarity responses and assigned the two least familiar topics; this approach maximised experimental control by ensuring uniformly low prior knowledge across participants. Participants were then offered a short break.

Before the search tasks, the researcher demonstrated the task flow using an example and provided a printed ISA scale reference.

³PsychoPy: <https://www.psychopy.org/>

⁴Lab Streaming Layer: <https://github.com/scn/labstreaminglayer>

The passive think-aloud protocol [75] was introduced via instructional video⁵. At the start of each task, the browser was set to Google's default search page. Following fNIRS headband positioning and a 30-second resting baseline recording, the search tasks commenced. All screen and audio activity was recorded, with events logged and synchronised via LSL (Fig. 4). The complete session lasted approximately one hour.

4 Results

4.1 Cognitive Style Grouping

Participants were classified into cognitive style groups based on the ratio of median reaction times to verbal versus imagery stimuli (VI dimension) and wholist versus analytic stimuli (WA dimension), following established procedures [82, 88]. The ratios on the two dimensions were uncorrelated ($r = 0.197$, $p > 0.05$), consistent with Riding's theory of orthogonal dimensions [87] and confirming their independence in our sample.

The mean VI ratio was 1.01 ($SD = 0.21$; range: 0.727–1.809), and the mean WA ratio was 1.11 ($SD = 0.25$; range: 0.741–2.198). Neither distribution satisfied normality assumptions (Shapiro-Wilk test, $p < 0.05$ for both) [93]. We therefore used median splits to classify participants, an approach recommended when sample distributions deviate from established population norms [82, 90]. Following the methodology of Riding et al. [90], sample medians ($Mdn_{VI} = 0.98$; $Mdn_{WA} = 1.06$) were used as cut-points to classify participants into four balanced groups: 20 verbalisers and 20 imagers, and 20 wholists and 20 analytics. The equal group sizes are a direct consequence of median-based classification with an even sample size, ensuring balanced statistical comparisons across dimensions.

4.2 Search Behaviours

The following analyses address RQ1, examining behavioural differences across cognitive style groups. To confirm that participants began with minimal domain knowledge, we evaluated the pre-search paragraphs. Of 40 participants, 34 reported no prior knowledge of the assigned topics, and the remaining 6 reported only superficial familiarity (e.g., “I have heard about them before but never took a deep look”). This confirmed uniformly low baseline knowledge across participants. Post-search paragraphs were not assessed for quality, as the primary purpose was to verify task engagement; all paragraphs were verified to contain task-relevant content, confirming that participants meaningfully engaged with the search task.

4.2.1 Information Search Strategy. Participants' search strategies were categorised into three approaches adapted from Kinley et al. [50]: *Top-down* (general-to-specific), *Bottom-up* (specific-to-general), and *No change* (consistent query scope throughout). Figure 5 illustrates examples of each pattern.

Top-down and *No change* strategies were the most prevalent; only one participant who belongs to the wholist-imagery group utilised a bottom-up approach. Due to insufficient frequency to satisfy chi-square assumptions, the *Bottom-up* category was excluded, and analyses focused on *Top-down* and *No change*. Chi-square tests [101] revealed no significant association between cognitive style and search strategy for either the VI dimension ($\chi^2(1) = 0.27$,

$p > 0.05$) or the WA dimension ($\chi^2(1) = 1.24$, $p > 0.05$). These findings contrast with Kinley et al. [50], who reported that verbalisers and wholists preferred *Top-down* approaches while imagers and analytics favoured *Bottom-up* strategies, but aligned with the findings of Hariri et al. [33]'s on the VI dimension.

4.2.2 Query Reformulation Behaviours. A Pearson correlation test [25] revealed a significant positive correlation between *Delete* and *Add* query behaviours ($r = 0.324$, $p < 0.05$), indicating that participants who removed content from queries also tended to add new content.

One-way ANOVAs (following verification of normality) examined differences across cognitive style groups for five reformulation behaviours. A significant difference emerged only for *Repeat* behaviour on the WA dimension ($F(1, 38) = 5.06$, $p < 0.05$): wholists *Repeated* previous queries significantly more often, whereas no analytic participants used this strategy. This finding aligns with Kinley et al. [50]. No significant differences were found between verbalisers and imagers for any reformulation behaviour.

4.2.3 Web Navigation Style. **Sporadic** searchers were characterised by a less systematic and more fragmented navigation pattern. They typically spent relatively longer time on query (re)formulation and then engaged in rapid browsing of SERP results. These participants tended to open multiple links in parallel, including both search results and internal hyperlinks, often switching frequently between tabs or windows. Their search process was marked by iterative movement between brief information encounters and repeated query reformulation, rather than sustained engagement with individual sources.

In contrast, **Structured** searchers exhibited a more organised and sequential search strategy. They carefully selected relevant links from the SERP and engaged in more sustained reading of the corresponding pages. Their navigation pattern followed a clearer step-by-step progression, where information gathered from a small number of carefully examined pages was used to progressively refine subsequent queries. Compared to Sporadic searchers, their browsing behaviour showed greater continuity and coherence, with fewer simultaneous open pages and more focused engagement with individual sources before returning to the query interface.

Chi-square tests revealed strong associations between cognitive style and SERP navigation preference. On the WA dimension ($\chi^2(1) = 12.13$, $p < 0.001$), 75% of analytics (15/20) preferred *Structured* navigation, whereas 80% of wholists (16/20) favoured *Sporadic* navigation. The analytic preference for *Structured* navigation aligns with Kinley et al. [50], though the wholist preference for *Sporadic* navigation contradicts their reported pattern.

On the VI dimension ($\chi^2(1) = 4.91$, $p < 0.05$), 65% of imagers (13/20) preferred *Structured* navigation, while 70% of verbalisers (14/20) favoured *Sporadic* navigation. This pattern is consistent with Kinley et al. [50] and Tang et al. [103], but conflicts with Graff [28] and Hariri et al. [33].

4.2.4 Information Processing Approach. **Reading** participants typically engaged in longer and more sustained interaction with page content, reflecting deeper information processing. In contrast, **Scanning** participants moved quickly through content, often spending brief periods (typically under 30 seconds) on a page before scrolling

⁵Think-aloud protocol training: <https://www.youtube.com/watch?v=g34tOmyKaMM>

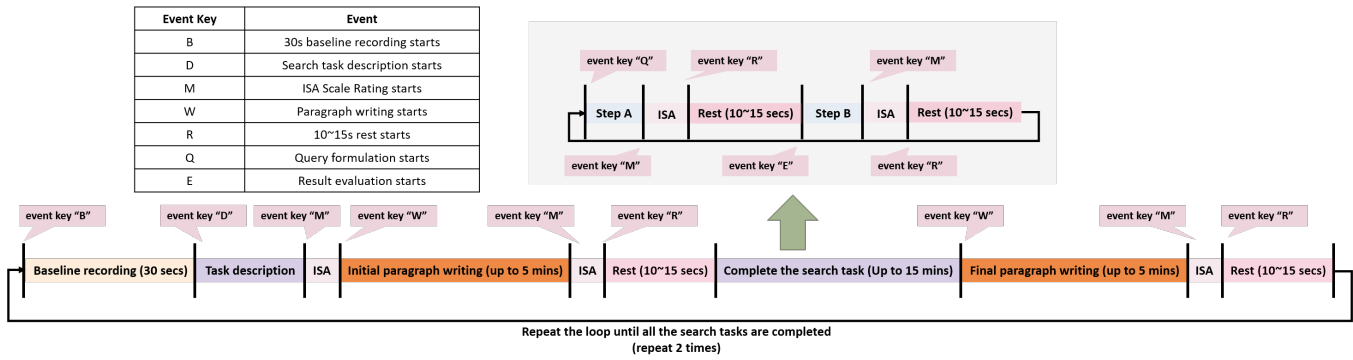


Figure 4: Event markers and task timeline for a single search task. Markers were synchronised with fNIRS recordings via Lab Streaming Layer, enabling precise alignment of behavioural events with fNIRS data. ISA = Instantaneous Self-Assessment.

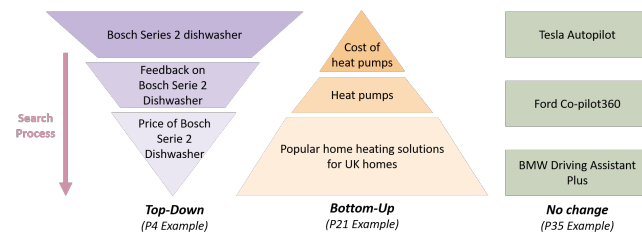


Figure 5: Examples of information search strategies observed across participants. Top-down: queries progress from general terms to specific products. Bottom-up: queries progress from specific items to broader categories. No change: queries maintain consistent specificity throughout the task.

or navigating to other results. In their think-aloud logs, they repeatedly reported their frustration with dense content reading, such as “this is too much to read” (Participant 38), “okay I will just read the titles” (Participant 42), and “let me have a look at the outline ... that is enough, move to the next page now” (Participant 51).

Chi-square tests revealed significant associations between cognitive style and information processing approach. The WA dimension showed a particularly strong relationship ($\chi^2(1) = 16.13, p < 0.001$): 90% of wholists (18/20) adopted *Scanning* strategies, whereas 65% of analytics (13/20) preferred *Reading*. A significant association also emerged for the VI dimension ($\chi^2(1) = 7.53, p < 0.05$): 80% of verbalisers (16/20) adopted *Scanning*, while 55% of imagers (11/20) preferred *Reading*. These patterns align with Kinley et al. [50] and Tang et al. [103], but contradict Hariri et al. [33]’s findings. Graff [28] did not report relevant results.

4.3 fNIRS Data

The following analyses address RQ2, examining brain activation patterns associated with the behavioural differences reported above. fNIRS data were not normally distributed (Shapiro-Wilk test, $p < 0.05$) [93]; Mann-Whitney U tests [65] were therefore used for statistical comparisons. P-values were adjusted using the False Discovery Rate (FDR) to control for Type I errors across multiple

channels and features, as recommended for fNIRS research [96]. The effect size was calculated according to Cohen [16].

4.3.1 Regions of Interest. Analyses focused on four regions of interest (ROIs) in the PFC: the dorsolateral prefrontal cortex (DLPFC) and ventrolateral prefrontal cortex (VLPFC) in both hemispheres. These regions were selected for their established roles in cognitive processes relevant to exploratory search. The DLPFC supports working memory [8], planning [17], and abstract reasoning [11]. The VLPFC is involved in memory retrieval [38], decision-making [37], and language processing [36, 37]. Channel-to-ROI mapping was performed using the fOLD toolbox (fNIRS Optodes Location Decider)⁶ [110].

4.3.2 Signal Processing. Signal processing was performed using MNE-Python (neuroimaging analysis software)⁷ [29] with the *artinis-brite23* montage. Preprocessing followed established protocols [84]: raw light intensity was converted to oxygenated (HbO) and deoxygenated (HbR) haemoglobin concentrations using the modified Beer-Lambert law [18]; motion artefacts were corrected using Temporal Derivative Distribution Repair [22]; data were bandpass filtered (0.01–0.1 Hz, fourth-order Butterworth); and short-separation channel signals were regressed from long-distance channels to remove systemic physiological noise [13].

Epochs were extracted using event markers for *Query Formulation* (Step A) and *Result Evaluation* (Step B), with baseline correction using the 10 seconds preceding each epoch onset (see Fig. 4). Each participant contributed at least three *Query Formulation* and two *Result Evaluation* epochs due to their varied epoch durations. Epoch lengths were set to 30 seconds for *Query Formulation* and 60 seconds for *Result Evaluation*, corresponding to the minimum completion times observed across participants for each stage; this approach ensures that all epochs contain complete task-related activity without truncation [62].

Given the varied task durations (30 seconds to 10 minutes), block averaging was used instead of general linear modelling (GLM). This approach is recommended for sustained cognitive tasks [92] and provides the same group-level results comparable to GLM for block-protocol designs [62].

⁶fOLD toolbox: <https://github.com/nirx/fOLD-public>

⁷MNE-Python: <https://mne.tools/stable/index.html>

Mean haemodynamic responses were calculated for each channel across epochs of the same type. To enhance statistical power, three features, listed in Table 2, were extracted from predefined ROIs.

Feature	Description
Area under the curve (AUC)	Cumulative signal amplitude across the epoch, indexing sustained brain activation [94] and overall mental workload [3].
Peak signal (5–7s)	Mean amplitude during the expected haemodynamic response peak window [107], capturing maximal haemodynamic response to task onset.
Initial slope (0–6s)	Slope of linear regression fitted to the first 6 seconds, reflecting response onset rate; sensitive to workload-related activation changes [64].

Table 2: fNIRS features extracted from each region of interest. Features capture complementary aspects of the haemodynamic response: cumulative activation (AUC), peak response magnitude, and response onset dynamics (slope).

4.3.3 Query Formulation Stage.

WA dimension. No significant channel-level differences emerged between analytics and wholists. However, at the ROI level, analytics showed consistently greater activation in the right VLPFC across all three features. AUC was significantly larger for analytics ($U = 325.00$, $p < 0.05$, *Cohen's* $r = 0.535$; $M = 0.141$) than wholists ($M = -0.217$). The initial slope was significantly steeper for analytics ($U = 281.00$, $p < 0.05$, *Cohen's* $r = 0.346$), indicating faster response onset. Peak signal was also significantly higher for analytics ($U = 285.00$, $p < 0.05$, *Cohen's* $r = 0.364$).

VI dimension. Imagers showed significantly higher HbO activation than verbalisers in channel S6_D5 (left DLPFC; $U = 277$, $p < 0.05$, *Cohen's* $r = 0.329$) and channel S8_D7 (left VLPFC; $U = 280$, $p < 0.05$, *Cohen's* $r = 0.342$). At the ROI level, imagers showed significantly greater left VLPFC activation for both AUC ($U = 297$, $p < 0.01$, *Cohen's* $r = 0.334$; imagers: $M = 0.685$; verbalisers: $M = -0.044$) and peak signal ($U = 287$, $p < 0.05$, *Cohen's* $r = 0.37$; imagers: $M = 0.079$; verbalisers: $M = -0.016$). The left DLPFC channel-level difference was not supported by ROI-level feature analysis and was therefore not interpreted further.

4.3.4 Result Evaluation Stage.

WA dimension. Channel-level analysis revealed significant differences between groups. Analytics showed greater HbO activation in channel S8_D7 (left VLPFC; $U = 275$, $p < 0.05$, *Cohen's* $r = 0.321$) and greater HbR in channel S6_D5 (left DLPFC; $U = 279$, $p < 0.05$, *Cohen's* $r = 0.338$). Wholists showed greater HbO activation in channel S6_D3 (right DLPFC; $U = 111$, $p < 0.05$, *Cohen's* $r = 0.381$).

At the ROI level, analytics exhibited significantly larger AUC in the left VLPFC ($U = 270.00$, $p < 0.05$, *Cohen's* $r = 0.299$), indicating sustained activation. Conversely, wholists showed significantly greater right DLPFC activation, with a higher slope ($U = 278.00$, $p < 0.05$, *Cohen's* $r = 0.334$) and peak signal ($U = 279.00$, $p < 0.05$, *Cohen's* $r = 0.338$). The left DLPFC channel-level finding lacked ROI-level support and was not interpreted further.

VI dimension. No significant channel-level differences emerged. However, imagers showed a significantly steeper HbO slope in the left VLPFC ($U = 291.00$, $p < 0.05$, *Cohen's* $r = 0.389$; imagers: $M = 0.002$; verbalisers: $M = -0.011$), indicating faster response onset.

4.4 Subjective Workload

ISA scores were compared across cognitive style groups for both the *Query Formulation* and the *Result Evaluation* stages using Mann-Whitney U tests, given the ordinal nature of the 5-point scale. No statistically significant differences for any comparison were found.

Descriptive trends suggested minor differences: wholists reported slightly higher workload than analytics during both *Query Formulation* ($M = 1.51$ vs. $M = 1.36$) and *Result Evaluation* ($M = 2.52$ vs. $M = 2.23$). On the VI dimension, verbalisers reported marginally higher workload during *Result Evaluation* ($M = 2.43$ vs. $M = 2.31$), while imagers reported marginally higher workload during *Query Formulation* ($M = 1.45$ vs. $M = 1.42$). These non-significant trends are interpreted cautiously in the Discussion.

5 Discussion

5.1 Main Findings

5.1.1 Cognitive Style and Search Behaviour (RQ1). Our findings demonstrate that cognitive style significantly predicts SERP interaction patterns and query refinement behaviours, while overall search strategy (*Top-down* vs. *Bottom-up*) did not differ across groups.

Participants predominantly adopted *Top-down* or *No change* strategies regardless of cognitive style, contrasting with Kinley et al. [50] who reported style-dependent strategy preferences, where verbalisers and wholists preferred a *Top-down* approach, while imagers and analytics preferred a *Bottom-up* strategy. This discrepancy likely reflects our task design: participants began with minimal domain knowledge, whereas real-world *Post-focus* searchers typically possess foundational understanding that may enable more varied strategic approaches. This suggests that the influence of cognitive style on high-level search strategy may be moderated by domain expertise—a relationship warranting future investigation.

On the WA dimension, wholists exhibited significantly more *Repeat* query behaviours and preferred *Sporadic* SERP navigation, rapidly switching between pages. Analytics, conversely, demonstrated *Structured* navigation and thorough *Reading* of results, processing information completely before proceeding—a pattern consistent with reduced need for query repetition.

On the VI dimension, despite no significant differences in *Query Formulation* behaviours, clear distinctions emerged in navigation and processing styles, which aligned with Tang et al. [103]'s findings. Verbalisers preferred *Sporadic* navigation with rapid *Scanning*, quickly evaluating multiple results. Imagers adopted *Structured* navigation with detailed *Reading*, systematically processing each result. These complementary patterns across dimensions highlight that cognitive style shapes not only what users do, but how they engage with search interfaces at multiple levels.

The significant positive correlation between *Delete* and *Add* query behaviours across all groups suggests that query refinement reflects active learning rather than search abandonment. Users dynamically adjust queries based on evolving understanding, a pattern consistent with the iterative nature of *Post-focus* information gathering [105].

These behavioural findings have practical implications for search interface design. Users who navigate *sporadically* and *scan* rapidly (wholists, verbalisers) may benefit from interfaces that preserve access to previously viewed content and present information densely.

Users who navigate *systematically* and *read* thoroughly (analytics, imagers) may benefit from richer result previews that support informed link selection.

5.1.2 Brain Activation and Behavioural Differences (RQ2). Our fNIRS data reveal distinct activation patterns in the PFC that correspond to the behavioural differences observed between cognitive style groups, providing neurophysiological evidence for the cognitive processes that may underlie style-related search behaviours.

Notably, subjective ISA ratings did not differ significantly across cognitive style groups, while fNIRS revealed clear distinctions. This divergence warrants careful interpretation. One possibility is that fNIRS captures latent cognitive effort, the objective cost of specific functions like semantic processing or navigation management, that may not be consciously perceived by the user. While self-reports reflect perceived MWL, fNIRS indexes the cortical resource allocation required by style, specific strategies in the PFC [34, 63]. The discrepancy underscores the value of multimodal measurement: self-report and neurophysiological measures may index different aspects of cognitive experience, and neither alone provides a complete picture.

WA dimension. Analytics showed higher right VLPFC activation during *Query Formulation*. This region is associated with inhibitory control [2] and deliberate decision-making under uncertainty [27], consistent with analytics' more careful, structured approach to query construction.

During *Result Evaluation*, analytics exhibited higher left VLPFC activation, a region implicated in semantic processing and controlled memory retrieval [7, 9]. This activation pattern is consistent with their preference for detailed *reading* and thorough information processing.

Wholists showed distinct activation in the right DLPFC during *Result Evaluation*, with a steeper initial response suggesting rapid engagement of executive resources. The right DLPFC supports working memory and cognitive flexibility [8], functions that would be heavily recruited during *Sporadic* navigation involving frequent task-switching. This high early demand may contribute to superficial initial processing, potentially explaining why wholists more frequently *Repeat* queries to revisit information.

VI dimension. Imagers showed significantly higher left VLPFC activation during both *Query Formulation* and initial *Result Evaluation*, despite no behavioural differences in *Query Formulation*. The left VLPFC supports language production and semantic retrieval [7, 9], suggesting that imagers expend greater effort translating visual representations into text-based queries. The steeper HbO slope during initial SERP exposure indicates intensive processing as imagers engage with the text-heavy interface. This cognitive demand likely promotes the more careful, systematic engagement reflected in their structured behavioural approach.

Our interpretations draw on established functional associations between PFC regions and cognitive processes [85]. While converging methods would strengthen causal claims, the alignment between activation patterns and observed behaviours provides coherent evidence for the cognitive demands imposed by different search strategies.

5.2 Implications for the IS Field

Our findings demonstrate that behavioural measures alone incompletely characterise cognitive style differences during exploratory search. While some behavioural differences were observed, the fNIRS data revealed additional activation differences even where no significant behavioural effects were detected (e.g., the VI dimension during *Query Formulation*). This suggests that observable interaction patterns capture only part of the underlying cognitive processes. This also helps explain inconsistencies in prior behavioural research [28, 33, 50, 103]: variability in reported findings may arise because the same underlying cognitive differences manifest differently depending on task characteristics and interface conditions. By providing converging neurophysiological evidence, our results strengthen the reliability of behaviour-based interpretations.

These findings carry direct implications for how cognitive style should be incorporated into personalised IS system design. A practical challenge is that search behaviours across cognitive style groups are largely similar at the early stages of an unfamiliar exploratory task, making it difficult to infer user characteristics from limited initial interaction data. As interaction data accumulates over time, however, behavioural differences may become more distinguishable. These observed patterns can be incorporated into long-term user models, allowing systems to infer a user's cognitive style and reuse this information across sessions. In this way, returning users can receive personalised support without requiring repeated pre-assessment. For first-time users with no prior interaction history, one practical approach is to obtain cognitive style traits through lightweight pre-assessment instruments (e.g., VICS or E-WA). This enables systems to provide personalised support from the outset, before sufficient behavioural evidence is available. Looking further ahead, the neural distinctions observed here point to a longer-term opportunity: as portable neuroimaging technologies develop, cognitive profiles may be inferred directly from neural signals, enabling implicit and real-time personalisation even before any interaction data is available.

For search interface design, our results suggest two complementary strategies:

Supporting effortful query formulation. Analytics and imagers both showed elevated prefrontal activation during query construction, consistent with deliberate, resource-intensive processing. Yet the source of that effort differs between them, and interfaces should respond accordingly.

For analytics, the MWL arises from attempting to construct a structured search plan and generate precise queries before sufficient background knowledge has been established. Interfaces could address this through *dynamic query suggestions* that help users decompose their current task into manageable sub-goals, relieving them of having to perform this decomposition independently. *Side-by-side result previews* that update as the query is typed can complement this by providing immediate feedback on query quality, supporting more informed reformulation decisions.

For imagers, the difficulty lies in translating a mental image into precise textual terms. Rather than requiring a fully formed query upfront, interfaces could offer a *node-based concept mapping* approach: users enter loosely associated keywords as nodes, and the system draws connections between them to surface relationships

and potential search directions. This allows iterative refinement across multiple cycles until the emerging query adequately reflects the user's intended meaning.

Accommodating diverse result-processing styles. Wholists and verbalisers, who tend to navigate sporadically and scan content rapidly, are best served by interfaces that reduce the cost of that scanning behaviour. Displaying more than 10 results per page with tighter spacing and bolded key terms lowers the effort of a rapid overview. A collapsible visited-links sidebar showing thumbnail previews of recently opened pages can further support their sporadic backtracking, and session-level history trails displayed alongside the SERP reduce reliance on browser navigation for content recovery.

Analytics and imagers, who process results more systematically and thoroughly, benefit instead from richer result representations. Extended abstracts, metadata (including publication dates and estimated reading times), and key-concept summaries allow these users to make more strategic decisions about which links to pursue, therefore ultimately reducing the need for exhaustive and detailed page-reading to assess a result's relevance.

5.3 Limitations and Future Directions

Several limitations constrain the generalisability of our findings.

First, our task decomposition, separating *Query Formulation* and *Result Evaluation* with enforced rest periods, departs from natural search behaviour, where these activities interleave fluidly. This structure was necessary to isolate fNIRS signals given haemodynamic latency, but may have altered the cognitive demands of the task. Findings may differ in unconstrained search, where users freely alternate between formulation and evaluation.

Second, our simulated *Post-focus* stage ensured controlled conditions but may not fully reflect search contexts where users possess substantial domain knowledge and pursue self-directed goals. However, the low-knowledge comparison task we employed is representative of common real-world scenarios—such as researching unfamiliar products, comparing options in new domains, or investigating topics outside one's expertise—where users must gather and synthesise information without prior grounding. The absence of strategy differences across cognitive styles may reflect this specific context, and different patterns might emerge when users search within their domains of expertise.

Third, as noted above, our activation-based interpretations rely on reverse inference from regional brain activation to cognitive function [85]. While we grounded interpretations in well-established functional associations, converging evidence from other methods (e.g., concurrent behavioural manipulation) would strengthen causal claims.

Fourth, the divergence between ISA and fNIRS measures raises questions about what each captures. Future work should systematically compare these measures to clarify their respective contributions to understanding search-related workload.

Future research should extend this work in several directions: examining other ISP stages with comparable neurophysiological detail; combining fNIRS with EEG to leverage their complementary temporal and spatial resolution; and validating design recommendations through controlled interface comparisons. While ecological validity remains important, controlled laboratory studies provide

essential foundations for understanding the cognitive processes underlying complex search behaviour.

6 Conclusion

Understanding the cognitive processes underlying individual differences in search behaviour is essential for designing adaptive information systems, yet the mechanisms by which cognitive style influences search remain poorly understood. This study addressed this gap by combining behavioural analysis with fNIRS neuroimaging to investigate how cognitive style shapes both observable behaviour and underlying brain activation during *Post-focus* exploratory search.

Addressing RQ1, our behavioural findings revealed that cognitive style significantly predicts SERP interaction patterns. Analytics and imagers consistently adopted *Structured* navigation and detailed *Reading* strategies, while wholists and verbalisers preferred *Sporadic* navigation with rapid *Scanning*. Query reformulation behaviours were largely consistent across groups, with the exception that wholists *Repeated* previous queries significantly more often than analytics.

Addressing RQ2, the fNIRS data provided neurophysiological evidence that corresponds to these behavioural patterns. Analytics showed elevated right VLPFC activation during *Query Formulation* and left VLPFC activation during *Result Evaluation*, consistent with deliberate decision-making and detailed semantic processing. Wholists exhibited heightened right DLPFC activation upon SERP exposure, consistent with increased executive demand from their *Sporadic* navigation style. Imagers showed greater left VLPFC activation than verbalisers during both *Query Formulation* and *Result Evaluation*, suggesting effortful translation between visual and verbal representations. Notably, these activation differences emerged even where behavioural measures revealed no significant distinctions, demonstrating the sensitivity of neurophysiological methods to cognitive processes that may not manifest in observable behaviour.

This work makes three contributions. First, it provides empirical evidence of behavioural differences between cognitive style groups during post-focus exploratory search. Second, by integrating neurophysiological measures, it shows that these differences are underpinned by distinct cognitive processes, thereby providing converging evidence that strengthens the validity of behavioural findings and addresses limitations of prior studies relying solely on behavioural observations, which led to inconsistent results. Third, it shows that fNIRS is a viable method for investigating cognitive processes in dynamic, interactive search tasks, supporting its use for studying complex information-seeking behaviour beyond static experimental settings.

These findings have direct implications for the design of personalised search interfaces that accommodate diverse cognitive styles, and establish a foundation for future research linking neurophysiology to complex information behaviour.

Acknowledgments

This work was supported by the 2025 Ningbo Yongjiang Talent – Educational Talent Cultivation Programme.

References

- [1] Chiara Acciarini, Federica Brunetta, and Paolo Boccadelli. 2021. Cognitive biases and decision-making strategies in times of change: a systematic literature review. *Management Decision* 59, 3 (2021), 638–652.
- [2] Nounagnon Frutueux Agbangla, Michel Audiffren, Jean Pylouster, and Cédric T Albinet. 2022. Load-dependent prefrontal cortex activation assessed by continuous-Wave Near-Infrared Spectroscopy during two executive tasks with three cognitive loads in young adults. *Brain Sciences* 12, 11 (2022), 1462.
- [3] Haleh Aghajani, Marc Garbey, and Ahmet Omurtag. 2017. Measuring mental workload with EEG+ fNIRS. *Frontiers in human neuroscience* 11 (2017), 359.
- [4] Hosam Al-Samarraie, Atef Eldenfria, and Husameddin Dawoud. 2017. The impact of personality traits on users' information-seeking behavior. *Information Processing & Management* 53, 1 (2017), 237–247.
- [5] Virgil L Anderson and Robert A McLean. 1974. *Design of experiments: a realistic approach*. Vol. 5. CRC Press.
- [6] Kumaripaba Athukorala, Dorota Glowacka, Giulio Jacucci, Antti Oulasvirta, and Jilles Vreeken. 2016. Is exploratory search different? A comparison of information search behavior for exploratory and lookup tasks. *Journal of the Association for Information Science and Technology* 67, 11 (2016), 2635–2651.
- [7] David Badre and Anthony D Wagner. 2007. Left ventrolateral prefrontal cortex and the cognitive control of memory. *Neuropsychologia* 45, 13 (2007), 2883–2901.
- [8] Aron K Barbey, Michael Koenigs, and Jordan Grafman. 2013. Dorsolateral prefrontal contributions to human working memory. *cortex* 49, 5 (2013), 1195–1205.
- [9] Sara Basso Moro, Simone Cutini, Maria Laura Ursini, Marco Ferrari, and Valentina Quaresima. 2013. Prefrontal cortex activation during story encoding/retrieval: a multi-channel functional near-infrared spectroscopy study. *Frontiers in Human Neuroscience* 7 (2013), 925.
- [10] Robert CA Bendall, Adam Galpin, Lynne P Marrow, and Simon Cassidy. 2016. Cognitive style: Time to experiment. *Frontiers in psychology* 7 (2016), 1786.
- [11] Silvia Bernardi, Marcus K Benna, Mattia Rigotti, Jérôme Munuera, Stefano Fusi, and C Daniel Salzman. 2020. The geometry of abstraction in the hippocampus and prefrontal cortex. *Cell* 183, 4 (2020), 954–967.
- [12] Siddharth Bhatt, Atahan Agrali, Kevin McCarthy, Rajneesh Suri, and Hasan Ayaz. 2019. Web usability testing with concurrent fNIRS and eye tracking. In *Neuroergonomics*. Elsevier, 181–186.
- [13] Sabrina Brigadoi and Robert J Cooper. 2015. How short is short? Optimum source–detector distance for short-separation channels in functional near-infrared spectroscopy. *Neurophotonics* 2, 2 (2015), 025005–025005.
- [14] Sherry Y Chen, George D Magoulas, and Dionisios Dimakopoulos. 2005. A flexible interface design for web directories to accommodate different cognitive styles. *Journal of the American Society for information science and Technology* 56, 1 (2005), 70–83.
- [15] Hong-Yu Cheng and Shu-Qiang Zhang. 2017. Examining the relationship between holistic/analytic style and classroom learning behaviors of high school students. *European journal of psychology of education* 32, 2 (2017), 271–288.
- [16] Jacob Cohen. 2013. *Statistical power analysis for the behavioral sciences*. routledge.
- [17] Cristiano Crescentini, Shima Seyed-Allaei, Antonino Vallesi, and Tim Shallice. 2012. Two networks involved in producing and realizing plans. *Neuropsychologia* 50, 7 (2012), 1521–1535.
- [18] David T Delpy, Mark Cope, Pieter van der Zee, SR Arridge, Susan Wray, and JS Wyatt. 1988. Estimation of optical pathlength through tissue from direct time of flight measurement. *Physics in Medicine & Biology* 33, 12 (1988), 1433.
- [19] Aude Dufresne and Sylvie Turcotte. 1997. Cognitive style and its implications for navigation strategies. *Artificial intelligence in education knowledge and media learning system* (1997), 287–293.
- [20] Susan T Dumais. 2016. Personalized Search: Potential and Pitfalls. In *Proceedings of the 25th ACM International Conference on Information and Knowledge Management (CIKM'16)*. 689.
- [21] Bruce Ferwerda, Li Chen, and Marko Tkalič. 2021. Psychological Models for Personalized Human-Computer Interaction (HCI). 673092 pages.
- [22] Frank A Fishburn, Ruth S Ludlum, Chandan J Vaidya, and Andrei V Medvedev. 2019. Temporal derivative distribution repair (TDDR): a motion correction method for fNIRS. *Neuroimage* 184 (2019), 171–179.
- [23] Frank A Fishburn, Megan E Norr, Andrei V Medvedev, and Chandan J Vaidya. 2014. Sensitivity of fNIRS to cognitive state and load. *Frontiers in human neuroscience* 8 (2014), 76.
- [24] Marsha E Fonteyn, Benjamin Kuipers, and Susan J Grobe. 1993. A description of think aloud method and protocol analysis. *Qualitative health research* 3, 4 (1993), 430–441.
- [25] David Freedman, Robert Pisani, and Roger Purves. 2007. *Statistics* (international student edition). *Pisani, R. Purves, 4th edn. WW Norton & Company*, New York (2007).
- [26] Naomi P Friedman and Trevor W Robbins. 2022. The role of prefrontal cortex in cognitive control and executive function. *Neuropsychopharmacology* 47, 1 (2022), 72–89.
- [27] Vinod Goel, Melanie Stollstorff, Marina Nakic, Kris Knutson, and Jordan Grafman. 2009. A role for right ventrolateral prefrontal cortex in reasoning about indeterminate relations. *Neuropsychologia* 47, 13 (2009), 2790–2797.
- [28] Martin Graff. 2005. Individual differences in hypertext browsing strategies. *Behaviour & Information Technology* 24, 2 (2005), 93–99.
- [29] Alexandre Gramfort, Martin Luessi, Eric Larson, Denis A. Engemann, Daniel Strohmeier, Christian Brodbeck, Roman Goj, Mainak Jas, Teon Brooks, Lauri Parkkonen, and Matti S. Hämäläinen. 2013. MEG and EEG Data Analysis with MNE-Python. *Frontiers in Neuroscience* 7, 267 (2013), 1–13. doi:10.3389/fnins.2013.00267
- [30] Jacek Gwizdzka. 2010. Distribution of cognitive load in web search. *Journal of the American Society for Information Science and Technology* 61, 11 (2010), 2167–2187.
- [31] Eugenia Hanfmann. 1941. A study of personal patterns in an intellectual performance. *Character & Personality: A Quarterly for Psychodiagnostic & Allied Studies* (1941).
- [32] Samyogita Hardikar, Brontë McKeown, Adam Turnbull, Ting Xu, Sofie L Valk, Boris C Bernhardt, Daniel S Margulies, Michael P Milham, Elizabeth Jefferies, Robert Leech, et al. 2024. Personality traits vary in their association with brain activity across situations. *Communications Biology* 7, 1 (2024), 1498.
- [33] Nadjla Hariri, Maryam Asadi, and Yazdan Mansourian. 2014. The impact of users' verbal/imagery cognitive styles on their Web search behavior. *Aslib Journal of Information Management* (2014).
- [34] Christian Herff, Dominic Heger, Ole Fortmann, Johannes Hennrich, Felix Putze, and Tanja Schultz. 2014. Mental workload during n-back task—quantified in the prefrontal cortex using fNIRS. *Frontiers in human neuroscience* 7 (2014), 935.
- [35] Gerard P Hodgkinson and Eugene Sadler-Smith. 2003. Complex or unitary? A critique and empirical re-assessment of the Allinson-Hayes Cognitive Style Index. *Journal of Occupational and Organizational Psychology* 76, 2 (2003), 243–268.
- [36] Paul Hoffman. 2019. Reductions in prefrontal activation predict off-topic utterances during speech production. *Nature communications* 10, 1 (2019), 515.
- [37] Juan Hui, Haiyue Dai, Lu Qi, Juan Wang, Guimei Cui, Junlin Mu, Lin Zhao, Shina Gu, Juan Li, and Zhao-Hui Zhang. 2025. Prefrontal activation in bipolar and unipolar depression patients in the letter fluency tasks and category fluency tasks: a functional near-infrared spectroscopy study. *Frontiers in Psychiatry* 16 (2025), 1610703.
- [38] Sahar Jahani, Antoniu L Fantana, David Harper, James M Ellison, David A Boas, Brent P Forester, and Meryem A Yücel. 2017. fNIRS can robustly measure brain activity during memory encoding and retrieval in healthy subjects. *Scientific reports* 7, 1 (2017), 9533.
- [39] Herbert H Jasper. 1958. The ten-twenty electrode system of the International Federation. *Electroencephalogr. Clin. Neurophysiol.* 10 (1958), 370–375.
- [40] Kaixin Ji, Danula Hettiachchi, Flora D Salim, Falk Scholer, and Damiano Spina. 2024. Characterizing information seeking processes with multiple physiological signals. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 1006–1017.
- [41] Oliver P John. 1990. The "Big Five" factor taxonomy: Dimensions of personality in the natural language and in questionnaires. *Handbook of personality: Theory and research* (1990).
- [42] CS Jordan and SD Brennen. 1992. Instantaneous self-assessment of workload technique (ISA). *Defence Research Agency, Portsmouth* (1992).
- [43] Diane Kelly and Jaime Teevan. 2003. Implicit feedback for inferring user preference: a bibliography. In *Acm Sigir Forum*, Vol. 37. ACM New York, NY, USA, 18–28.
- [44] Kyung-Sun Kim. 2001. Information-seeking on the Web: Effects of user and task variables. *Library & information science research* 23, 3 (2001), 233–255.
- [45] Yong-Mi Kim and Soo Young Rieh. 2005. Dual-task performance as a measure of mental effort in searching a library system and the web. *Proceedings of the American Society for Information Science and Technology* 42, 1 (2005).
- [46] Khamsum Kinley and Dian Tjondronegoro. 2010. The impact of users' cognitive style on their navigational behaviors in web searching. In *Proceedings of the 15th Australasian Document Computing Symposium*. School of Computer Science and IT, RMIT University, 68–75.
- [47] Khamsum Kinley, Dian Tjondronegoro, and Helen Partridge. 2010. Web searching interaction model based on user cognitive styles. In *Proceedings of the 22nd Conference of the Computer-Human Interaction Special Interest Group of Australia on Computer-Human Interaction*. 340–343.
- [48] Khamsum Kinley, Dian Tjondronegoro, Helen Partridge, and Sylvia Edwards. 2012. Human-computer interaction: the impact of users' cognitive styles on query reformulation behaviour during web searching. In *Proceedings of the 24th Australasian Computer-Human Interaction Conference*. 299–307.
- [49] Khamsum Kinley, Dian Tjondronegoro, Helen Partridge, and Sylvia Edwards. 2012. Relationship between the nature of the search task types and query reformulation behaviour. In *Proceedings of the Seventeenth Australasian Document Computing Symposium*. 39–46.
- [50] Khamsum Kinley, Dian Tjondronegoro, Helen Partridge, and Sylvia Edwards. 2014. Modeling users' web search behavior and their cognitive styles. *Journal of*

- the Association for Information Science and Technology 65, 6 (2014), 1107–1123.
- [51] Marta Koć-Januchta, Tim Höfler, Gun-Brit Thoma, Helmut Prechtel, and Detlev Leutner. 2017. Visualizers versus verbalizers: Effects of cognitive style on learning with texts and pictures—An eye-tracking study. *Computers in Human Behavior* 68 (2017), 170–179.
 - [52] Maria Kozhevnikov, Carol Evans, and Stephen M Kosslyn. 2014. Cognitive style as environmentally sensitive individual differences in cognition: A modern synthesis and applications in education, business, and management. *Psychological science in the public interest* 15, 1 (2014), 3–33.
 - [53] David JM Kraemer, Lauren M Rosenberg, and Sharon L Thompson-Schill. 2009. The neural correlates of visual and verbal cognitive styles. *Journal of Neuroscience* 29, 12 (2009), 3792–3798.
 - [54] Carol C Kuhlthau. 1993. seeking Meaning: A Process Approach. *Library and Information Science. Norwood, NJ: Ablex* (1993).
 - [55] Kemal Lazović, Filipe Madeira, Eftim Zdravetski, Luis Augusto Silva, Paulo Jorge Coelho, and Ivan Miguel Pires. 2025. An analytical review of optimization techniques in information retrieval for enhanced decision support. *Decision Analytics Journal* (2025), 100657.
 - [56] Thomas Le Bras, Benoit Allibe, and Karine Doré-Mazars. 2024. The way we look at an image or a webpage can reveal personality traits. *Scientific Reports* 14, 1 (2024), 15488.
 - [57] Theresa René LeBlanc. 2018. Learning styles: academic fact or urban myth? A recent review of the literature. *Journal of college academic support programs* 1, 1 (2018), 7–7.
 - [58] Jingjing Liu and Nicholas J Belkin. 2015. Personalizing information retrieval for multi-session tasks: Examining the roles of task stage, task type, and topic knowledge on the interpretation of dwell time as an indicator of document usefulness. *Journal of the Association for Information Science and Technology* 66, 1 (2015), 58–81.
 - [59] Jingjing Liu, Chang Liu, and Nicholas J Belkin. 2020. Personalization in text information retrieval: A survey. *Journal of the Association for Information Science and Technology* 71, 3 (2020), 349–369.
 - [60] Jia-Jiunn Lo and Yun-Jay Wang. 2012. Development of an adaptive EC website with online identified cognitive styles of anonymous customers. *International Journal of Human-Computer Interaction* 28, 9 (2012), 560–575.
 - [61] Kristiyan Lukanov, Horia A Maior, and Max L Wilson. 2016. Using fNIRS in usability testing: understanding the effect of web form layout on mental workload. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 4011–4016.
 - [62] Robert Luke, Eric Larson, Maureen J Shader, Hamish Innes-Brown, Lindsey Van Yper, Adrian KC Lee, Paul F Sowman, and David McAlpine. 2021. Analysis methods for measuring passive auditory fNIRS responses generated by a block-design paradigm. *Neurophotonics* 8, 2 (2021), 025008–025008.
 - [63] Horia A Maior, Max L Wilson, and Sarah Sharples. 2018. Workload alerts—using physiological measures of mental workload to provide feedback during tasks. *ACM Transactions on Computer-Human Interaction (TOCHI)* 25, 2 (2018), 1–30.
 - [64] Kevin Mandrick, Gérard Derosiere, Gérard Dray, Denis Coulon, Jean-Paul Micallef, and Stéphane Perrey. 2013. Utilizing slope method as an alternative data analysis for functional near-infrared spectroscopy-derived cerebral hemodynamic responses. *International Journal of Industrial Ergonomics* 43, 4 (2013), 335–341.
 - [65] Henry B Mann and Donald R Whitney. 1947. On a test of whether one of two random variables is stochastically larger than the other. *The annals of mathematical statistics* (1947), 50–60.
 - [66] Alan Medlar, Denis Kotkov, and Dorota Glowacka. 2024. Unexplored Frontiers: A Review of Empirical Studies of Exploratory Search. *SIGIR Forum* 58, 1 (Aug. 2024), 1–19. doi:10.1145/3687273.3687278
 - [67] S Messick. 1976. Personality difference in cognition and creativity. In (S. Messick, Ed.), *Individuality in learning*. London: Jossey-Bass (1976).
 - [68] Christian Meurisch, Cristina A Mihale-Wilson, Adrian Hawlitschek, Florian Giger, Florian Müller, Oliver Hinz, and Max Mühlhäuser. 2020. Exploring user expectations of proactive AI systems. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 4 (2020), 1–22.
 - [69] Alan Miller. 1987. Cognitive styles: An integrated model. *Educational psychology* 7, 4 (1987), 251–268.
 - [70] Yashar Moshfeghi, Peter Triantafillou, and Frank Pollick. 2019. Towards predicting a realisation of an information need based on brain signals. In *The world wide web conference*. 1300–1309.
 - [71] Yashar Moshfeghi, Peter Triantafillou, and Frank E Pollick. 2016. Understanding information need: An fMRI study. In *Proceedings of the 39th International ACM SIGIR conference on Research and Development in Information Retrieval*. 335–344.
 - [72] Robert S Moyer and Richard H Bayer. 1976. Mental comparison and the symbolic distance effect. *Cognitive Psychology* 8, 2 (1976), 228–246.
 - [73] Ajaya Neupane, Nitesh Saxena, and Leanne Hirshfield. 2017. Neural underpinnings of website legitimacy and familiarity detection: An fNIRS study. In *Proceedings of the 26th International Conference on World Wide Web*. 1571–1580.
 - [74] Heather L O'Brien, Rebecca Dickinson, and Nicole Askin. 2017. A scoping review of individual differences in information seeking behavior and retrieval research between 2000 and 2015. *Library & Information Science Research* 39, 3 (2017), 244–254.
 - [75] Erica L Olmsted-Hawala, Elizabeth D Murphy, Sam Hawala, and Kathleen T Ashenfelter. 2010. Think-aloud protocols: a comparison of three think-aloud protocols for use in testing data-dissemination web sites for usability. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 2381–2390.
 - [76] Dan Pan, Yijing Zhang, Zhizhong Li, and Zhiqiang Tian. 2021. Effects of cognitive characteristics and information format on teleoperation performance: A cognitive fit perspective. *International Journal of Industrial Ergonomics* 84 (2021), 103157.
 - [77] Jonathan Peirce, Jeremy R Gray, Sol Simpson, Michael MacAskill, Richard Höchenberger, Hiroyuki Sogo, Erik Kastman, and Jonas Kristoffer Lindeløv. 2019. PsychoPy2: Experiments in behavior made easy. *Behavior research methods* 51, 1 (2019), 195–203.
 - [78] Yingzhe Peng, Xiaoting Qin, Zhiyang Zhang, Jue Zhang, Qingwei Lin, Xu Yang, Dongmei Zhang, Saravan Rajmohan, and Qi Zhang. 2025. Navigating the Unknown: A Chat-Based Collaborative Interface for Personalized Exploratory Tasks. In *Proceedings of the 30th International Conference on Intelligent User Interfaces*. 1048–1063.
 - [79] Elizabeth R Peterson, Ian J Deary, and Elizabeth J Austin. 2003. On the assessment of cognitive style: four red herrings. *Personality and Individual Differences* 34, 5 (2003), 899–904.
 - [80] Elizabeth R Peterson, Ian J Deary, and Elizabeth J Austin. 2003. The reliability of Riding's cognitive style analysis test. *Personality and individual differences* 34, 5 (2003), 881–891.
 - [81] Elizabeth R Peterson, Ian J Deary, and Elizabeth J Austin. 2005. Are intelligence and personality related to verbal-imagery and wholistic-analytic cognitive styles? *Personality and Individual Differences* 39, 1 (2005), 201–213.
 - [82] Elizabeth R Peterson, Ian J Deary, and Elizabeth J Austin. 2005. A new measure of Verbal-Imagery Cognitive Style: VICS. *Personality and Individual Differences* 38, 6 (2005), 1269–1281.
 - [83] Matthew F Pike, Horia A Maior, Martin Porcheron, Sarah C Sharples, and Max L Wilson. 2014. Measuring the effect of think aloud protocols on workload using fNIRS. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 3807–3816.
 - [84] Paola Pinti, Felix Scholkmann, Antonia Hamilton, Paul Burgess, and Ilias Tachtsidis. 2019. Current status and issues regarding pre-processing of fNIRS neuroimaging data: an investigation of diverse signal filtering methods within a general linear model framework. *Frontiers in human neuroscience* 12 (2019), 505.
 - [85] Russell A. Poldrack. 2006. Can cognitive processes be inferred from neuroimaging data? *Trends in Cognitive Sciences* 10, 2 (2006), 59–63. doi:10.1016/j.tics.2005.12.004
 - [86] Ali Reza Rezaei and Larry Katz. 2004. Evaluation of the reliability and validity of the cognitive styles analysis. *Personality and Individual Differences* 36, 6 (2004), 1317–1327.
 - [87] Richard Riding and Indra Cheema. 1991. Cognitive styles—an overview and integration. *Educational psychology* 11, 3–4 (1991), 193–215.
 - [88] Richard J Riding. 1991. Cognitive styles analysis—CSA administration. *Birmingham: Learning and Training and Technology* (1991).
 - [89] Richard J Riding. 1997. On the nature of cognitive style. *Educational psychology* 17, 1–2 (1997), 29–49.
 - [90] Richard J Riding, Alan Glass, Stuart R Butler, and Christopher W Pleydell-Pearce. 1997. Cognitive style and individual differences in EEG alpha during information processing. *Educational Psychology* 17, 1–2 (1997), 219–234.
 - [91] Richard J Riding, Alan Glass, and Graeme Douglas. 1993. Individual differences in thinking: Cognitive and neurophysiological perspectives. *Educational Psychology* 13, 3–4 (1993), 267–279.
 - [92] Hendrik Santosa, Frank Fishburn, Xuetong Zhai, and Theodore J Huppert. 2019. Investigation of the sensitivity-specificity of canonical-and deconvolution-based linear models in evoked functional near-infrared spectroscopy. *Neurophotonics* 6, 2 (2019), 025009–025009.
 - [93] Samuel Sanford Shapiro and Martin B Wilk. 1965. An analysis of variance test for normality (complete samples). *Biometrika* 52, 3–4 (1965), 591–611.
 - [94] Tripp Shealy, John Gero, Mo Hu, and Julie Milovanovic. 2020. Concept generation techniques change patterns of brain activation during engineering design. *Design Science* 6 (2020), e31.
 - [95] Md Hedayetul Islam Shovon, D Nandagopal, Jia Tina Du, Ramasamy Vijayalakshmi, and Bernadine Cocks. 2015. Cognitive activity during web search. In *Proceedings of the 38th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 967–970.
 - [96] Archana K Singh and Ipeita Dan. 2006. Exploring the false discovery rate in multichannel NIRS. *Neuroimage* 33, 2 (2006), 542–549.
 - [97] Ben Steichen and Bo Fu. 2019. Towards adaptive information visualization—a study of information visualization aids and the role of user cognitive style. *Frontiers in artificial intelligence* 2 (2019), 22.
 - [98] Tristan Stenner, Chadwick Boulay, Matthew Grivich, David Medine, Christian Kothe, tobiasherzke, Giso Grimm, xloem, Arthur Biancarelli, Boris Mansencal, chauser, Jérémy Frey, kyucrane, Samuel Powell, Pierre Clisson, and phfix. 2022.

- sccn/libls: v1.16.0. (March 2022). doi:10.5281/zenodo.6387090
- [99] Alistair Sutcliffe and Mark Ennis. 1998. Towards a cognitive theory of information retrieval. *Interacting with computers* 10, 3 (1998), 321–351.
 - [100] Alistair G. Sutcliffe, Mark Ennis, and SJ Watkinson. 2000. Empirical studies of end-user information searching. *Journal of the American society for information science* 51, 13 (2000), 1211–1231.
 - [101] Ronald J Tallarida and Rodney B Murray. 1987. Chi-square test. In *Manual of pharmacologic calculations: with computer programs*. Springer, 140–142.
 - [102] Desney S Tan and Anton Nijholt. 2010. In *Brain-computer interfaces*. Springer-Verlag London Limited, Chapter Brain-Computer Interfaces and Human-Computer Interaction, 8–9.
 - [103] Huimin Tang, Johann Benerradi, Horia A Maior, Matthew Pike, Aleksandra Landowska, and Max L Wilson. 2024. Exploring the Impact of Verbal-Imagery Cognitive Style on Web Search Behaviour and Mental Workload. In *Proceedings of the 2024 Conference on Human Information Interaction and Retrieval*. 303–316.
 - [104] Mohammed Rashid ul Ansar and MP Ganesh. 2023. Cognitive Styles Research: A Review of Theory, Assessments and Applications. *International Journal of Indian Psychology* 11, 1 (2023), 924–934.
 - [105] Pertti Vakkari. 2001. A theory of the task-based information retrieval process: A summary and generalisation of a longitudinal study. *Journal of documentation* (2001).
 - [106] Arno Villringer, J Planck, C Hock, L Schleinkofer, and U Dirnagl. 1993. Near infrared spectroscopy (NIRS): a new tool to study hemodynamic changes during activation of brain function in human adults. *Neuroscience letters* 154, 1-2 (1993), 101–104.
 - [107] Pramudi Wijayasiri, Douglas EH Hartley, and Ian M Wiggins. 2017. Brain activity underlying the recovery of meaning from degraded speech: A functional near-infrared spectroscopy (fNIRS) study. *Hearing research* 351 (2017), 55–67.
 - [108] Ziyi Ye, Xiaohui Xie, Yiqun Liu, Zhihong Wang, Xuancheng Li, Jiaji Li, Xuesong Chen, Min Zhang, and Shaoping Ma. 2022. Why Don't You Click: Understanding Non-Click Results in Web Search with Brain Signals. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 633–645.
 - [109] Xiaojun Yuan, Xiangman Zhang, Chaomei Chen, and Joshua M Avery. 2011. Seeking information with an information visualization system: a study of cognitive styles. *Information research: an international electronic journal* 16, 4 (2011), n4.
 - [110] Guilherme Augusto Zimeo Morais, Joana Bisol Balardin, and João Ricardo Sato. 2018. fNIRS Optodes' Location Decider (FOLD): a toolbox for probe arrangement guided by brain regions-of-interest. *Scientific reports* 8, 1 (2018), 3341.